



Figure 3: This is how a video looks after embedding it within `<video>` tags

and CSS. This enables Web developers to build video player interfaces that are more consistent with the rest of their website. The ability to control playback with JavaScript allows video to be a more native part of the user experience in interactive Web applications.

This has given way to an absolutely dedicated HTML tag called `<video>`. The `<video>` tag defines video, such as a movie clip or other video streams. All one needs to do is embed a URL to a video file and the video gets embedded into the page, all by itself. The various attributes available for the `<video>` tags are reproduced in Table 2.

Explaining Ogg Theora and H.264 codecs

Ogg Theora is an open format that is thought to be unencumbered by patents. The primary reference implementation is distributed under an open source licence and is being developed by the non-profit Xiph.org with funding from Mozilla. Ogg is strongly preferred by the FOSS community because it can be freely redistributed without requiring licensing fees.

H.264 is a high-performance codec that is maintained by the ISO/IEC Moving Picture Experts Group (MPEG) as part of the MPEG-4 family. It is emerging as the dominant codec for both streaming video and optical media, as it is said to deliver the visual quality of MPEG-2 (used on DVDs) at roughly half the bit-rate. The MPEG LA consortium manages licensing of the underlying patents that cover

H.264 compression algorithms and other software methods needed to implement the codec. In order to use the format, adopters have to pay licensing fees to MPEG LA.

Debates over patents on video codecs

Patent encumbrance is one of the driving forces behind the HTML 5 video codec controversy. The patent licensing requirements mean that H.264 codecs can't be freely redistributed, making the format a non-starter for Mozilla and most other open source browser vendors. Opera also objects, saying that the licensing fees are too high. Mozilla and Opera strongly advocate Ogg Theora as an alternative because its freedom from known patents could ensure that there are no licensing barriers that prevent ubiquitous adoption.

Apple objects to Ogg Theora, claiming that the lack of known patents on Theora doesn't rule out the threat of submarine patents that could eventually be used against adopters. Apple is also concerned about the lack of widespread support for hardware-based Theora decoding, a factor that diminishes the format's viability on mobile devices. Google shares Apple's scepticism about the potential of Theora in the marketplace. The search giant claims that Theora's lack of quality relative to H.264 will make it an impractical choice for large-scale streaming video services such as YouTube. (For the complete e-mail sent by Ian Hickson, visit <http://lists.whatwg.org/htdig.cgi/whatwg-whatwg.org/2009-June/020620.html>).

A solution that seems logical on the surface is to simply expose each platform's underlying media playback engine through the HTML 5 video element—DirectShow on Windows, GStreamer on Linux, and QTKit on Mac OS X. This would make it possible for the browser to play any video formats that are supported natively on the user's computer.

Now, there's a push for hardware

decoding that makes Theora on mobiles technically possible and working well. If Apple is satisfied on the legal aspects and jumps on board, that changes the game. I think Google is mostly ambivalent since it supports both right now. Opera doesn't want H.264 anyway. IE 8 can likely be handled by a plug-in. Apple really is the deciding factor. So, Theora seems to be, pretty much, the future.

Audio interface

It is just as simple to embed audio into a page using the audio element. Most of the attributes are common between the video and audio elements, although for obvious reasons, the audio element lacks the width, height, and poster attributes.

```
<audio src="audio.oga" controls>
  <a href="audio.oga">get the track</a>
</audio>
```

HTML 5 provides the source element to specify alternative video and audio files that the browser may choose from, based on its media type or codec support. The media attribute can be used to specify a media query for selection based on the device limitations and the type attribute for specifying the media type and codecs. Note that when using the source elements, the src attribute needs to be omitted from the parent video or audio element, or the alternatives given by the source elements will be ignored.

```
<video poster="VidPost.jpg">
  <source src="vid.3gp" type="video/3gpp">
  media="handheld">
  <source src="vid.ogv" type="video/ogg;
  codecs=theora, vorbis">
  <source src="vid.mp4" type="video/mp4">
</video>
<audio>
  <source src="audio.oga" type="audio/ogg">
  <source src="audio.mp3" type="audio/mpeg">
</audio>
```

For authors who want a little more control over the user interface so that they can make it fit the overall design